

HACKATHON 2026

Real-Time Portfolio Risk & Concentration Alert System

*Monitors live portfolio positions and flags concentration and risk-limit breaches in real time
— with Claude-generated rationale behind every alert.*

Duration	Saturday 10 AM – Sunday 4 PM (July 11 – 12, 2026 approx. 31 hrs)
Team Size	3 – 4 members per team
LLM Stack	Anthropic Claude API (one key provided per team)
Domain	Financial Services – Portfolio Risk Management

OVERVIEW

Portfolio managers and risk desks must continuously monitor thousands of positions across accounts and funds for concentration breaches — single-issuer exposure, sector overweights, geography limits, and correlated risk clusters. Manual review is periodic and reactive, letting breaches persist for hours or days before they are caught. Teams will build an AI-powered platform that ingests live or batch portfolio holdings, uses Claude to evaluate exposure against configurable risk and concentration limits, classifies the severity of any breach, generates a clear rationale, and automatically alerts the right desk — all with a complete audit trail.

CORE DELIVERABLES

Task 1 – Portfolio Data Ingestion	Build a system that accepts portfolio holdings and position data (equities, bonds, derivatives, cash) across multiple accounts/funds, in real-time or batch feed format, and normalizes it for downstream analysis.
Task 2 – AI-Powered Concentration & Exposure Analysis	Use the Anthropic Claude API to evaluate each portfolio against configurable concentration limits (single issuer, sector, geography, asset class) and correlation/volatility signals, identifying and explaining any limit breach.
Task 3 – Risk Severity Scoring	Produce a structured alert verdict — LOW, MEDIUM, HIGH, or CRITICAL — with a confidence score. Claude must generate a human-readable rationale explaining the exposure factors that drove the classification.
Task 4 – Automated Escalation & Notifications	Based on the severity verdict, automatically trigger at least two downstream actions to escalate the alert to the correct desk and notify relevant stakeholders.

EXAMPLE SYSTEM OUTPUT

Portfolio ID: PORT-2026-0442 Submitted: 2026-07-11 09:15:32 IST Fund: Alpha Growth Opportunities Fund
Type: Multi-Asset – Long Only
Exposure Analysis (Claude):
Issuer Concentration : Reliance Industries Ltd — 9.8% of NAV [BREACH X] (limit: 8.0%)
Sector Concentration : Energy — 22.4% of NAV [WARNING] (limit: 25.0%, approaching threshold)
Geography Concentration : India — 61% of NAV [OK ✓] (limit: 70%)
Correlation Cluster : 3 holdings show >0.85 rolling 30-day correlation [FLAGGED]
Risk Assessment (Claude): Breach Severity: HIGH (single-issuer limit breached by 1.8 pts)
Volatility Context : Reliance 30-day realized volatility up 40% quarter-over-quarter
Historical Pattern : similar breaches in Q1 preceded a 2-day rebalancing lag
Overall Verdict: HIGH RISK — IMMEDIATE REBALANCING REVIEW
Confidence: 91% Estimated Review Time: ~15 min
Automated Actions Triggered:
1. Alert pushed to Risk Desk Slack channel — #portfolio-risk-alerts
2. Rebalancing ticket created — Jira: RISK-3312 assigned to Portfolio Manager
3. Daily risk dashboard updated with breach flag and audit-trail entry

EVALUATION CRITERIA

Criteria	Description	Weight
AI Exposure Analysis & Rationale	Accuracy of Claude-driven concentration analysis, limit-breach detection, and rationale quality	25%
Automation & Escalation	Effectiveness of post-assessment escalation workflows and notification actions	20%
Risk Model Quality	Sensibility of concentration/risk scoring model; handling of edge cases and conflicting signals	15%
Working Demo	Live end-to-end demo: ingest positions → analyze exposure → score → alert, using provided sample data	25%
API Efficiency	Purposeful Claude API usage; prompt design and token cost management	10%
Docs / README	Markdown documentation quality, setup instructions, and system architecture clarity	5%